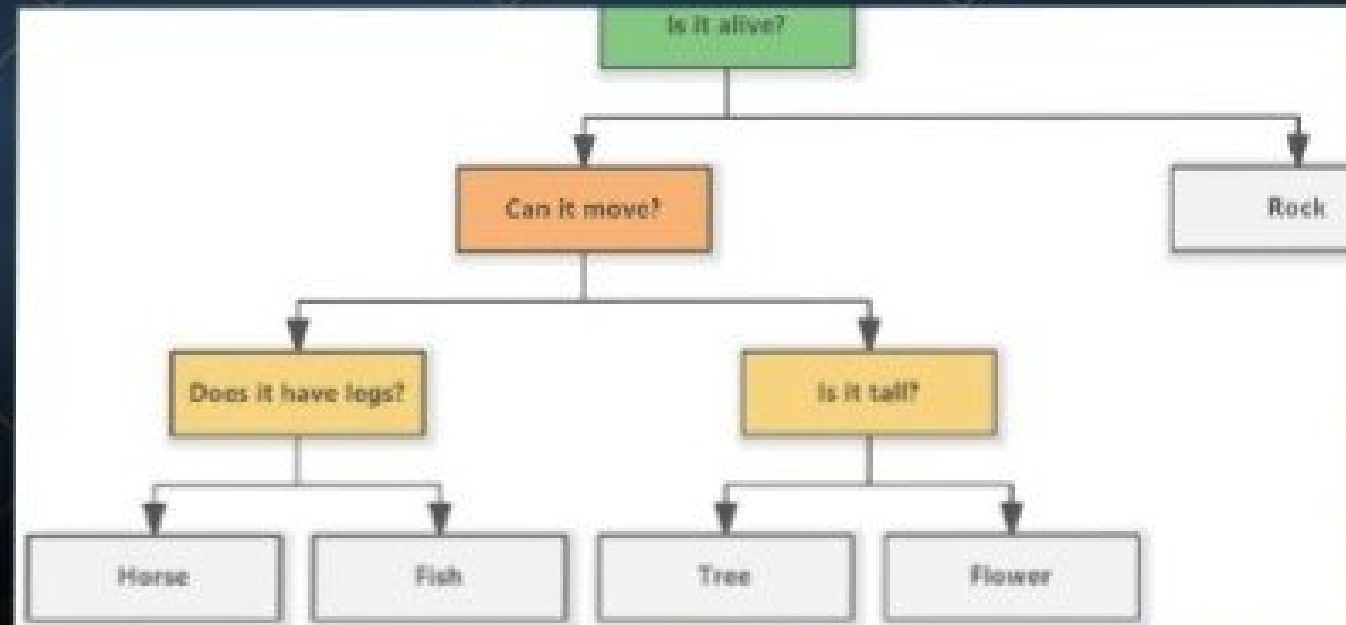
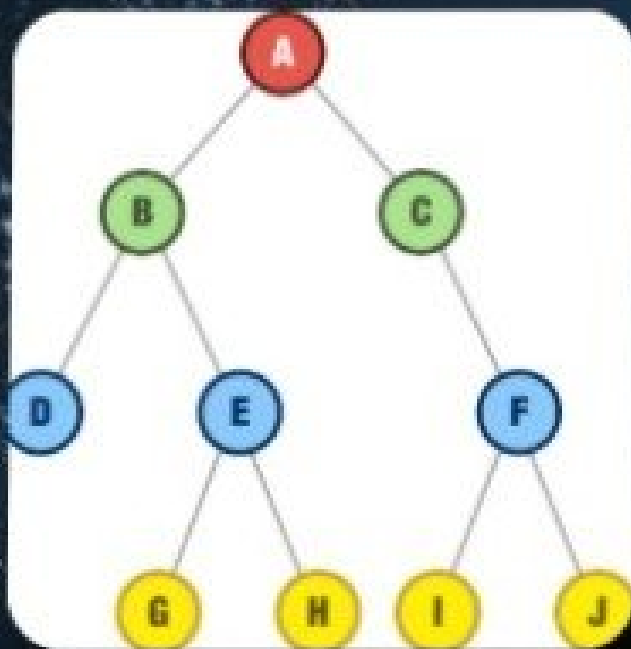


Vide Game AI - Decision Tree CRUD for NPC Behavior

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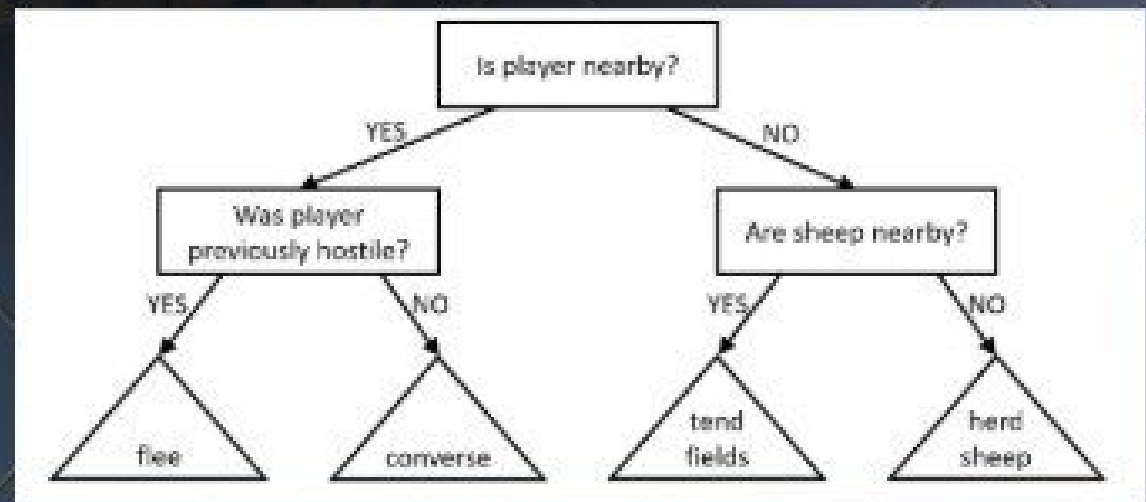


- Data Structure Used
- Binary Tree
- Node contains:
 - ID
 - Decision
 - Left child
 - Right child

● Node Structure (Code)

```
struct Node
{
    int id;
    char decision[100];
    struct Node *left;
    struct Node *right;
};
```

● **System Design**
Example Decision Tree:
Enemy Nearby?
Yes → Attack Enemy
No → Patrol Area





Problem Statement

NPCs need smart decision-making

Manual logic is complex

Need system to:

Add decisions

Update decisions

Delete decisions

Search decisions

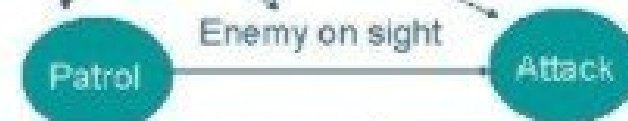
Solution: Decision Tree + CRUD Operation

Decision Tree

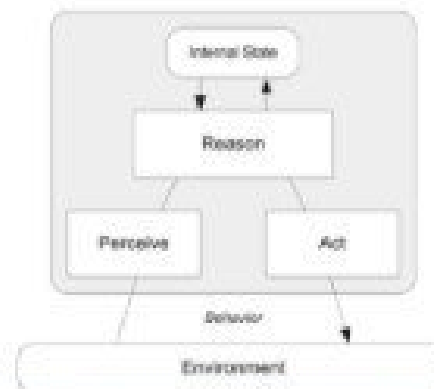


NPC's Behavior is Modeled through Finite State Machines (I)

- **State**: an activity performed by an avatar
- **Event**: something that happens in the game world that makes state change



NPC Behavior



Introduction

- **NPCs (Non-Player Characters) make decisions in games**
- **Actions include:**
 - **Attack**
 - **Patrol**
 - **Run**
 - **Search**
- **Decision Trees help simulate intelligent behavior**

● Algorithm (Search)

1. Compare ID with root
2. If equal → Found
3. If smaller → Left
4. If larger → Right

● Algorithm (Update & Delete)

Update:

- Search node
- Modify decision

Delete:

- Find node
- Remove it
- Adjust tree

● CRUD Operations

- Create → Add Node
- Read → Display Tree
- Update → Modify Decision
- Delete → Remove Node

● Algorithm (Insert)

1. Start
2. Check if tree empty
3. Create node
4. Compare IDs
5. Insert left or right



Implementation Details

C Programming

Structures

malloc & free

Functions

Tree traversal



Menu Operations

Add Node

Delete Node

Update Node

Search Node

Display Tree

Exit

● Sample Output

===== VIDEO GAME AI =====

1. Add Node
2. Display Tree

Node ID: 10

Decision: Attack Enemy

● Advantages

Simple decision making

Easy to modify

Efficient structure

Real-world application

● Limitations

- Console-based only
- No graphical view
- Basic AI logic

● Future Enhancements

- Graphical tree visualization
- Game engine integration
- Advanced AI logic
- Multiple conditions

Conclusion

- Implemented Decision Tree
- CRUD operations successful
- Useful for Game AI
- Learned data structures & memory

Thank you